MA104 · DISCRETE MATHEMATICS

Assignment 01 — Truth Tables and Propositional Logic

Questions: **36**
Due: **09 / 10 / 2026**

Chapter 1 — The Foundations: Logic and Proofs

Name: ..... **ID:** .....

Group: ..... **Date:** .....

Instructions. Work through the parts in order; each part builds on the one before it.

- Fill in **every** cell. A truth table must show all 2^n rows and all the intermediate columns printed for you — a final column on its own is not enough.
- Use **T** and **F** only (no 1 / 0, no ticks).
- Where a question asks for a conclusion — valid, satisfiable, holds, and so on — write it on the ruled line and name the row that justifies it.
- Hand-write your answers on this sheet. Write your name, ID and group above.
- **Answer without AI.** ChatGPT, Claude, Gemini and similar tools are not allowed, and neither is copying from a classmate. You may be asked to explain any answer at the board, and an answer you cannot explain will not be accepted.

Notation

Symbol	Name	Read as
$\neg p$	negation	not p
$p \wedge q$	conjunction	p and q
$p \vee q$	disjunction	p or q (or both)
$p \oplus q$	exclusive or	p or q, but not both
$p \to q$	conditional	if p then q
$p \leftrightarrow q$	biconditional	p if and only if q
$\varphi \vDash \psi$	logical consequence	ψ follows from φ
$\varphi \equiv \psi$	logical equivalence	φ and ψ have the same truth table

Part A — Truth Tables of the Basic Connectives

Fifteen short tables over the two variables p and q.

Question 1

Complete the truth table for:

$$p \wedge q$$

p	q	$p \wedge q$
T	T	_____
T	F	_____
F	T	_____
F	F	_____

Question 2

Complete the truth table for:

$$p \vee q$$

p	q	$p \vee q$
T	T	_____
T	F	_____
F	T	_____
F	F	_____

Question 3

Complete the truth table for:

$$p \to q$$

p	q	$p \to q$
T	T	_____
T	F	_____
F	T	_____
F	F	_____

Question 4

Complete the truth table for:

$$p \leftrightarrow q$$

p	q	$p \leftrightarrow q$
T	T	_____
T	F	_____
F	T	_____
F	F	_____

Question 5

Complete the truth table for:

$$\neg p \vee q$$

p	q	$\neg p$	$\neg p \vee q$
T	T	_____	_____
T	F	_____	_____
F	T	_____	_____
F	F	_____	_____

Question 6

Complete the truth table for:

$$p \wedge \neg q$$

p	q	$\neg q$	$p \wedge \neg q$
T	T	_____	_____
T	F	_____	_____
F	T	_____	_____
F	F	_____	_____

Question 7

Complete the truth table for:

$$\neg(p \wedge q)$$

p	q	$p \wedge q$	$\neg(p \wedge q)$
T	T	_____	_____
T	F	_____	_____
F	T	_____	_____
F	F	_____	_____

Question 8

Complete the truth table for:

$$\neg(p \vee q)$$

p	q	$p \vee q$	$\neg(p \vee q)$
T	T	_____	_____
T	F	_____	_____
F	T	_____	_____
F	F	_____	_____

Question 9

Complete the truth table for:

$$(p \wedge q) \to p$$

p	q	$p \wedge q$	$(p \wedge q) \to p$
T	T	_____	_____
T	F	_____	_____
F	T	_____	_____
F	F	_____	_____

Question 10

Complete the truth table for:

$$p \to (q \vee p)$$

p	q	$q \vee p$	$p \to (q \vee p)$
T	T	_____	_____
T	F	_____	_____
F	T	_____	_____
F	F	_____	_____

Question 11

Complete the truth table for:

$$(p \to q) \wedge p$$

p	q	$p \to q$	$(p \to q) \wedge p$
T	T	_____	_____
T	F	_____	_____
F	T	_____	_____
F	F	_____	_____

Question 12

Complete the truth table for:

$$(p \vee q) \wedge \neg p$$

p	q	$p \vee q$	$\neg p$	$(p \vee q) \wedge \neg p$
T	T	_____	_____	_____
T	F	_____	_____	_____
F	T	_____	_____	_____
F	F	_____	_____	_____

Question 13

Complete the truth table for:

$$(p \wedge q) \vee \neg q$$

p	q	$p \wedge q$	$\neg q$	$(p \wedge q) \vee \neg q$
T	T	_____	_____	_____
T	F	_____	_____	_____
F	T	_____	_____	_____
F	F	_____	_____	_____

Question 14

Complete the truth table for:

$$(p \to q) \leftrightarrow (\neg q \to \neg p)$$

p	q	$p \to q$	$\neg q$	$\neg p$	$\neg q \to \neg p$	**Final**
T	T	_____	_____	_____	_____	_____
T	F	_____	_____	_____	_____	_____
F	T	_____	_____	_____	_____	_____
F	F	_____	_____	_____	_____	_____

Question 15

Complete the truth table for:

$$(p \vee q) \to (p \wedge q)$$

p	q	$p \vee q$	$p \wedge q$	$(p \vee q) \to (p \wedge q)$
T	T	_____	_____	_____
T	F	_____	_____	_____
F	T	_____	_____	_____
F	F	_____	_____	_____

Part B — Exclusive Or

*$p \oplus q$ is true exactly when p and q have **different** truth values, and false when they agree. Where a second final column is printed, compare the two and say whether they match.*

Question 16

Complete the truth table for:

$$p \oplus q$$

p	q	$p \oplus q$
T	T	_____
T	F	_____
F	T	_____
F	F	_____

Question 17

Complete the truth table for:

$$\neg(p \oplus q)$$

p	q	$p \oplus q$	$\neg(p \oplus q)$
T	T	_____	_____
T	F	_____	_____
F	T	_____	_____
F	F	_____	_____

Question 18

Complete the truth table for:

$$(p \vee q) \wedge \neg(p \wedge q)$$

p	q	$p \vee q$	$p \wedge q$	$\neg(p \wedge q)$	Final
T	T	_____	_____	_____	_____
T	F	_____	_____	_____	_____
F	T	_____	_____	_____	_____
F	F	_____	_____	_____	_____

Question 19

Complete the truth table for:

$$(p \wedge \neg q) \vee (\neg p \wedge q)$$

p	q	$\neg q$	$p \wedge \neg q$	$\neg p$	$\neg p \wedge q$	Final
T	T	_____	_____	_____	_____	_____
T	F	_____	_____	_____	_____	_____
F	T	_____	_____	_____	_____	_____
F	F	_____	_____	_____	_____	_____

Question 20

Complete the truth table for:

$$\neg p \oplus q$$

p	q	$\neg p$	$\neg p \oplus q$	$\neg(p \oplus q)$
T	T	_____	_____	_____
T	F	_____	_____	_____
F	T	_____	_____	_____
F	F	_____	_____	_____

Question 21

Complete the truth table for:

$$\neg p \oplus \neg q$$

p	q	$\neg p$	$\neg q$	$\neg p \oplus \neg q$	$p \oplus q$
T	T	_____	_____	_____	_____
T	F	_____	_____	_____	_____
F	T	_____	_____	_____	_____
F	F	_____	_____	_____	_____

Question 22

Complete the truth table for:

$$(p \oplus q) \oplus q$$

p	q	$p \oplus q$	Final
T	T	_____	_____
T	F	_____	_____
F	T	_____	_____
F	F	_____	_____

Question 23

Complete the truth table for:

$$(p \oplus q) \to (p \vee q)$$

p	q	$p \oplus q$	$p \vee q$	Final
T	T	_____	_____	_____
T	F	_____	_____	_____
F	T	_____	_____	_____
F	F	_____	_____	_____

Question 24

Complete the truth table for:

$$p \oplus p \qquad p \oplus \neg p$$

p	$\neg p$	$p \oplus p$	$p \oplus \neg p$
T	_____	_____	_____
F	_____	_____	_____

Question 25

Associativity. Complete both final columns and say whether $(p \oplus q) \oplus r$ and $p \oplus (q \oplus r)$ always agree.

$$(p \oplus q) \oplus r \qquad p \oplus (q \oplus r)$$

p	q	r	$p \oplus q$	$(p \oplus q) \oplus r$	$q \oplus r$	$p \oplus (q \oplus r)$
T	T	T	_____	_____	_____	_____
T	T	F	_____	_____	_____	_____
T	F	T	_____	_____	_____	_____
T	F	F	_____	_____	_____	_____
F	T	T	_____	_____	_____	_____
F	T	F	_____	_____	_____	_____
F	F	T	_____	_____	_____	_____
F	F	F	_____	_____	_____	_____

Conclusion: .....

Question 26

Parity. The claim is that $p \oplus q \oplus r$ is true exactly when an **odd** number of p, q, r are true. Complete the table and say whether the claim holds.

$$p \oplus q \oplus r$$

p	q	r	Number of T's	$p \oplus q \oplus r$	Odd?
T	T	T	_____	_____	_____
T	T	F	_____	_____	_____
T	F	T	_____	_____	_____
T	F	F	_____	_____	_____
F	T	T	_____	_____	_____
F	T	F	_____	_____	_____
F	F	T	_____	_____	_____
F	F	F	_____	_____	_____

Conclusion: .....

Part C — Well-Formed Formulas and Interpretations

Before computing with a formula, you must know that it is a formula at all, and what an interpretation does to it.

Question 27 Ex. 2.1

Which of the following are well formed propositional formulas? For each one write **WFF** or **not a WFF**; if it is not, say which rule of the grammar it violates.

1. $\vee pq$ .....
2. $(\neg(p \to (q \wedge p)))$ .....
3. $(\neg(p \to (q = p)))$ .....
4. $(\neg(\blacklozenge\, (q \vee p)))$ .....
5. $(p \wedge \neg q) \vee (q \to r)$ .....
6. $p\neg r$ .....

Question 28 Ex. 2.2

Let v be the interpretation with $v(p) = F$, $v(q) = T$, $v(r) = T$. Does v satisfy the following formulas? Give the value of each formula under v, then answer **yes** or **no**.

Formula	Value under v	Satisfied?
$(p \to \neg q) \vee \neg(r \wedge q)$	_____	_____
$(\neg p \vee \neg q) \to (p \vee \neg r)$	_____	_____
$\neg(\neg p \to \neg q) \wedge r$	_____	_____
$\neg(\neg p \to q \wedge \neg r)$	_____	_____

Part D — Validity, Satisfiability, Consequence and Equivalence

Use the truth-table method throughout. State the conclusion on the ruled line and name the row that settles it.

Question 29 Ex. 2.3

Compute the truth table of $(F \vee G) \wedge \neg(F \wedge G)$.

F	G	$F \vee G$	$F \wedge G$	$\neg(F \wedge G)$	Final
T	T	_____	_____	_____	_____
T	F	_____	_____	_____	_____
F	T	_____	_____	_____	_____
F	F	_____	_____	_____	_____

Note: .....

Question 30 Ex. 2.4

Use the truth-table method to determine whether $(p \to q) \vee (p \to \neg q)$ is **valid**.

p	q	$\neg q$	$p \to q$	$p \to \neg q$	Final
T	T	_____	_____	_____	_____
T	F	_____	_____	_____	_____
F	T	_____	_____	_____	_____
F	F	_____	_____	_____	_____

Conclusion: .....

Question 31 Ex. 2.5

Use the truth-table method to determine whether $\varphi : (\neg p \vee q) \wedge (q \to \neg r \wedge \neg p) \wedge (p \vee r)$ is **satisfiable**.

p	q	r	$\neg p \vee q$	$\neg r \wedge \neg p$	$q \to \neg r \wedge \neg p$	$p \vee r$	φ
T	T	T	_____	_____	_____	_____	_____
T	T	F	_____	_____	_____	_____	_____
T	F	T	_____	_____	_____	_____	_____
T	F	F	_____	_____	_____	_____	_____
F	T	T	_____	_____	_____	_____	_____
F	T	F	_____	_____	_____	_____	_____
F	F	T	_____	_____	_____	_____	_____
F	F	F	_____	_____	_____	_____	_____

Conclusion: .....

Question 32 Ex. 2.6

Use the truth-table method to determine whether $\varphi : p \wedge \neg q \to p \wedge q$ is a **logical consequence** of $\psi : \neg p$.

p	q	$\psi : \neg p$	$p \wedge \neg q$	$p \wedge q$	φ
T	T	_____	_____	_____	_____
T	F	_____	_____	_____	_____
F	T	_____	_____	_____	_____
F	F	_____	_____	_____	_____

Conclusion: .....

Question 33 Ex. 2.7

Use the truth-table method to determine whether $p \to (q \wedge \neg q)$ and $\neg p$ are **logically equivalent**.

p	q	$q \wedge \neg q$	$p \to (q \wedge \neg q)$	$\neg p$	Equal?
T	T	_____	_____	_____	_____
T	F	_____	_____	_____	_____
F	T	_____	_____	_____	_____
F	F	_____	_____	_____	_____

Conclusion: .....

Part E — Extended Truth-Table Practice

Three long questions, each with several lettered formulas.

Question 34 Ex. 2.8

Compute the truth tables of the following propositional formulas.

(a) $(p \to p) \to p$

p	$p \to p$	Final
T	_____	_____
F	_____	_____

(b) $p \to (p \to p)$

p	$p \to p$	Final
T	_____	_____
F	_____	_____

(c) $p \vee q \to p \wedge q$

p	q	$p \vee q$	$p \wedge q$	Final
T	T	_____	_____	_____
T	F	_____	_____	_____
F	T	_____	_____	_____
F	F	_____	_____	_____

(d) $p \vee (q \wedge r) \to (p \wedge r) \vee q$

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \wedge r$	$(p \wedge r) \vee q$	Final
T	T	T	_____	_____	_____	_____	_____
T	T	F	_____	_____	_____	_____	_____
T	F	T	_____	_____	_____	_____	_____
T	F	F	_____	_____	_____	_____	_____
F	T	T	_____	_____	_____	_____	_____
F	T	F	_____	_____	_____	_____	_____
F	F	T	_____	_____	_____	_____	_____
F	F	F	_____	_____	_____	_____	_____

(e) $p \to (q \to p)$

p	q	$q \to p$	Final
T	T	_____	_____
T	F	_____	_____
F	T	_____	_____
F	F	_____	_____

(f) $(p \wedge \neg q) \vee \neg(p \leftrightarrow q)$

p	q	$\neg q$	$p \wedge \neg q$	$p \leftrightarrow q$	$\neg(p \leftrightarrow q)$	Final
T	T					
T	F					
F	T					
F	F					

Question 35 Ex. 2.9

For each formula below, use the truth-table method to decide whether it is **valid**, **satisfiable** or **unsatisfiable**, and write the verdict on the line under the table.

(a) $(p \to q) \wedge \neg q \to \neg p$

p	q	$p \to q$	$\neg q$	$(p \to q) \wedge \neg q$	$\neg p$	Final
T	T					
T	F					
F	T					
F	F					

Verdict: .....

(b) $(p \to q) \to (p \to \neg q)$

p	q	$p \to q$	$\neg q$	$p \to \neg q$	Final
T	T				
T	F				
F	T				
F	F				

Verdict: .....

(c) $(p \vee q \to r) \vee p \vee q$

p	q	r	$p \vee q$	$p \vee q \to r$	Final
T	T	T			
T	T	F			
T	F	T			
T	F	F			
F	T	T			
F	T	F			
F	F	T			
F	F	F			

Verdict: .....

(d) $(p \vee q) \wedge (p \to r \wedge q) \wedge (q \to \neg r \wedge p)$

p	q	r	$p \vee q$	$r \wedge q$	$\neg r \wedge p$	$p \to r \wedge q$	$q \to \neg r \wedge p$	Final
T	T	T						
T	T	F						
T	F	T						
T	F	F						
F	T	T						
F	T	F						
F	F	T						
F	F	F						

Verdict: .....

(e) $(p \to (q \to r)) \to ((p \to q) \to (p \to r))$

p	q	r	$q \to r$	$p \to (q \to r)$	$p \to q$	$p \to r$	$(p \to q) \to (p \to r)$	Final
T	T	T						
T	T	F						
T	F	T						
T	F	F						
F	T	T						
F	T	F						
F	F	T						
F	F	F						

Verdict: .....

(f) $(p \vee q) \wedge (\neg q \wedge \neg p)$

p	q	$p \vee q$	$\neg q \wedge \neg p$	Final
T	T			
T	F			
F	T			
F	F			

Verdict: .....

(g) $(\neg p \to q) \vee ((p \wedge \neg r) \leftrightarrow q)$

p	q	r	$\neg p \to q$	$p \wedge \neg r$	$(p \wedge \neg r) \leftrightarrow q$	Final
T	T	T				
T	T	F				
T	F	T				
T	F	F				
F	T	T				
F	T	F				
F	F	T				
F	F	F				

Verdict: .....

(h) $(p \to q) \wedge (p \to \neg q)$

p	q	$p \to q$	$p \to \neg q$	Final
T	T			
T	F			
F	T			
F	F			

Verdict: .....

(i) $(p \to (q \vee r)) \vee (r \to \neg p)$

p	q	r	$q \vee r$	$p \to (q \vee r)$	$r \to \neg p$	Final
T	T	T				
T	T	F				
T	F	T				
T	F	F				
F	T	T				
F	T	F				
F	F	T				
F	F	F				

Verdict: .....

Question 36 Ex. 2.10

Use the truth-table method to check the following logical consequences and equivalences. Compare the two final columns of each table, then write **holds** or give an interpretation that refutes the claim.

(a) $(p \to q) \vDash \neg p \to \neg q$

p	q	$p \to q$	$\neg p$	$\neg q$	$\neg p \to \neg q$
T	T				
T	F				
F	T				
F	F				

Holds? .....

(b) $(p \to q) \land \neg q \vDash \neg p$

p	q	$p \to q$	$\neg q$	$(p \to q) \land \neg q$	$\neg p$
T	T				
T	F				
F	T				
F	F				

Holds? .....

(c) $p \to q \land r \vDash (p \to q) \to r$

p	q	r	$q \land r$	$p \to q \land r$	$p \to q$	$(p \to q) \to r$
T	T	T				
T	T	F				
T	F	T				
T	F	F				
F	T	T				
F	T	F				
F	F	T				
F	F	F				

Holds? .....

(d) $p \lor (\neg q \land r) \vDash q \lor \neg r \to p$

p	q	r	$\neg q \land r$	$p \lor (\neg q \land r)$	$q \lor \neg r$	$q \lor \neg r \to p$
T	T	T				
T	T	F				
T	F	T				
T	F	F				
F	T	T				
F	T	F				
F	F	T				
F	F	F				

Holds? .....

(e) $\neg(p \wedge q) \equiv \neg p \vee \neg q$

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p \vee \neg q$
T	T	_____	_____	_____
T	F	_____	_____	_____
F	T	_____	_____	_____
F	F	_____	_____	_____

Holds? .....

(f) $(p \vee q) \wedge (\neg p \to \neg q) \equiv q$

p	q	$p \vee q$	$\neg p \to \neg q$	LHS	q
T	T	_____	_____	_____	_____
T	F	_____	_____	_____	_____
F	T	_____	_____	_____	_____
F	F	_____	_____	_____	_____

Holds? .....

(g) $(p \wedge q) \vee r \equiv (p \to \neg q) \to r$

p	q	r	$p \wedge q$	$(p \wedge q) \vee r$	$p \to \neg q$	$(p \to \neg q) \to r$
T	T	T	_____	_____	_____	_____
T	T	F	_____	_____	_____	_____
T	F	T	_____	_____	_____	_____
T	F	F	_____	_____	_____	_____
F	T	T	_____	_____	_____	_____
F	T	F	_____	_____	_____	_____
F	F	T	_____	_____	_____	_____
F	F	F	_____	_____	_____	_____

Holds? .....

(h) $(p \vee q) \wedge (\neg p \to \neg q) \equiv p$

p	q	$p \vee q$	$\neg p \to \neg q$	LHS	p
T	T	_____	_____	_____	_____
T	F	_____	_____	_____	_____
F	T	_____	_____	_____	_____
F	F	_____	_____	_____	_____

Holds? .....

(i) $((p \to q) \to q) \to q \equiv p \to q$

p	q	$p \to q$	$(p \to q) \to q$	LHS	$p \to q$
T	T	_____	_____	_____	_____
T	F	_____	_____	_____	_____
F	T	_____	_____	_____	_____
F	F	_____	_____	_____	_____

Holds? .....

— END OF ASSIGNMENT —